

Task 1: Hidden Markov Model for Human Activity Recognition

Problem Statement

A wearable fitness tracker is used to monitor a person's physical activity based on sensor readings. The actual activity being performed is *hidden*, while the wearable records the intensity of movement during fixed time intervals. The movement intensity is categorized as *Low*, *Medium*, or *High*.

The Hidden Markov Model (HMM) is defined as follows.

- **Hidden States:**

$$S = \{\text{Walking, Running, Resting}\}$$

- **Observation Symbols:**

$$V = \{\text{Low, Medium, High}\}$$

- **Initial State Probability Vector:**

$$\pi = [0.5 \quad 0.3 \quad 0.2]$$

- **State Transition Probability Matrix:**

$$A = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.3 & 0.2 & 0.5 \end{bmatrix}$$

where each row represents the current hidden state and each column represents the next hidden state.

- **Emission Probability Matrix:**

$$B = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.1 & 0.3 & 0.6 \\ 0.7 & 0.2 & 0.1 \end{bmatrix}$$

where each row represents a hidden state (Walking, Running, Resting) and each column represents an observation symbol (Low, Medium, High).

- **Observation Sequence:**

$$O = (\text{High, Medium, Low, High})$$

Tasks

1. Compute the forward probabilities for each time step using the Forward Algorithm.
2. Construct the complete Forward Probability (α) table.
3. Determine the likelihood of the given observation sequence.
4. Implement the Forward Algorithm in Python and verify the computed likelihood.

Task 2: Human Activity Recognition using `hmmlearn`

Problem Statement

Using the Hidden Markov Model parameters provided in Task 1, implement the HMM using the `hmmlearn` Python library.

Tasks

1. Install the `hmmlearn` library and import the required modules.
2. Construct a Categorical Hidden Markov Model using the given initial state probabilities, transition probability matrix, and emission probability matrix.
3. Compute the likelihood of the observation sequence

$$O = (\text{High}, \text{Medium}, \text{Low}, \text{High}).$$

4. Compute the likelihood for each of the following observation sequences:
 - $O_1 = (\text{Low}, \text{Medium}, \text{High}, \text{Medium})$
 - $O_2 = (\text{High}, \text{High}, \text{High}, \text{High})$
 - $O_3 = (\text{Low}, \text{Low}, \text{Low}, \text{Low})$
 - $O_4 = (\text{Medium}, \text{Medium}, \text{Medium}, \text{Medium})$
5. Briefly explain the relationship between the observation likelihood computed using the Forward Algorithm and the likelihood obtained using `hmmlearn`.